BIOCHEMISTRY CUMULATIVE EXAM
MARCH 9, 2001

1. When researchers attempt to clone and express the gene for a protein the manipulation of the nucleic acid often proceeds smoothly. The gene is in frame with good promoters and so on. At the nucleic acid level everything seems perfect yet sometimes no functional protein can be expressed. This is especially true when genes from eukaryotes are expressed in prokaryotes, but not limited to this situation. State briefly two reasons why protein expression can fail in this way. (40 points)

2. Explain how to solve one of the two problems you note above. (20 points)

3. Often the next problem is purifying an expressed protein. Developing a method of purification for a specific protein is often laborious and may be very difficult. Several general methods have been developed that work quite well to purify almost any protein that has successfully cloned and expressed. Explain how one of these methods works and the steps necessary in cloning, expression, and/or purification to carry it out. (25 points)

4. Name three proteins that have been successfully cloned, expressed, and purified for commercial or medical purposes (proteins cloned exclusively for research use do not count). Explain their commercial value briefly. (15 points)

ANALYTICAL CUMULATIVE EXAMINATION
MARCH 9, 2001

This famous cartoon by James Gillray shows some experiments performed at the Royal Institution in which newly discovered gases were inhaled. On March 22, 1800, after visiting such a lecture, Lady Holland wrote:

> "This Institution of Rumford's furnishes ridiculous stories. The other day they tried the effect of the gas, so poetically described by Beddoes; it exhilarates the spirits, distends the machine. The first subject was a corpulent middle-aged gentleman, who after inhaling a sufficient dose, was requested to describe to the company his sensations; 'Why, I only feel stupid'. This intelligence was received amidst a burst of applause, most probably not for the novelty of the information. Sir Coxe Hippisley was the next who submitted to the operation, but the effect upon him was so animating that the ladies tittered, held up their hands, and declared themselves satisfied."

1. What is the gas being inhaled? (10%)

2. How could you demonstrate the identity and composition of that gas:

    a. Using techniques available in the early 1800's? (30%)

    b. Using modern techniques? (60%)

Describe in detail the methods used, giving and explaining the relevant equations. Credit will be given for the use of numerical examples.